

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

# AWS re:Invent

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# Advanced networking on Amazon S3

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# Agenda

Introduction to networking for Amazon S3

Accessing Amazon S3 using VPC endpoints

AWS PrivateLink for Amazon S3

S3 Multi-Region Access Points

Recent launches

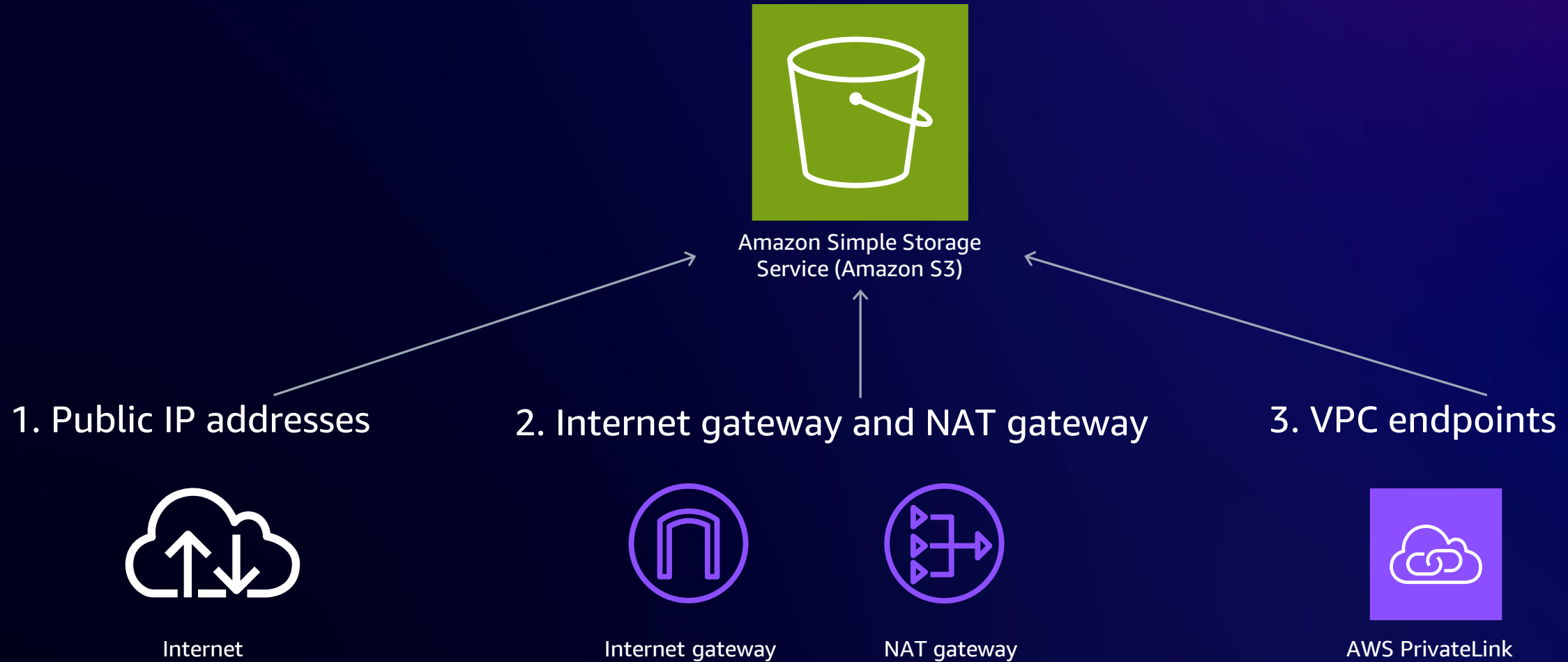
Advanced use cases

Recap



# Introduction to networking for Amazon S3

# Introduction to networking for Amazon S3



# VPC endpoints for Amazon S3

# VPC endpoints for Amazon S3



Secure access to  
Amazon S3 over the  
Amazon network



Easy to configure,  
reliable, and  
highly secure



Control access with  
endpoint policies

## Types of VPC endpoints to access Amazon S3

1. Gateway endpoints

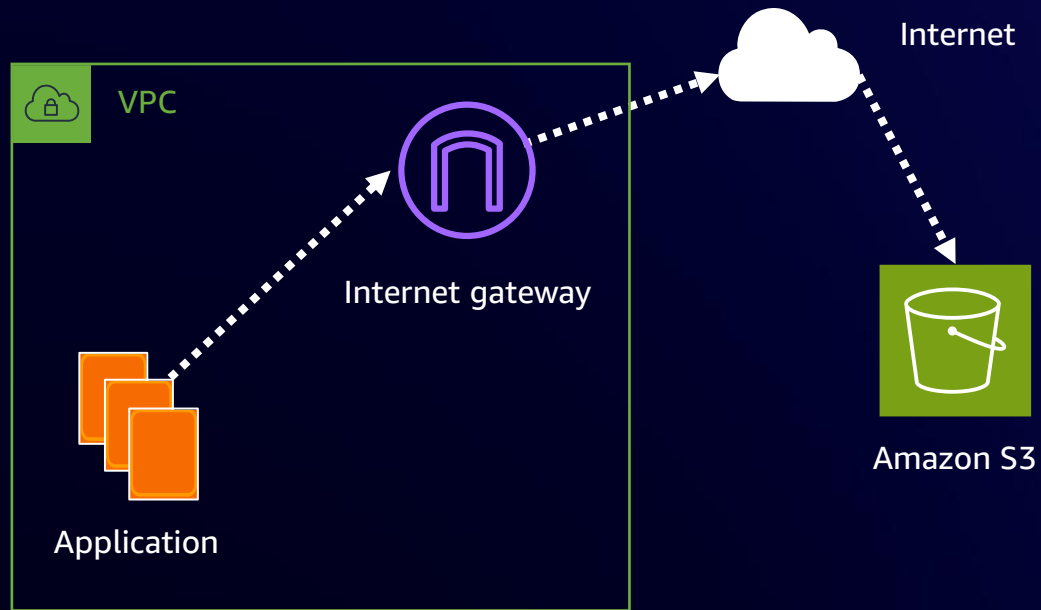
2. Interface endpoints

# Accessing Amazon S3 using VPC endpoints

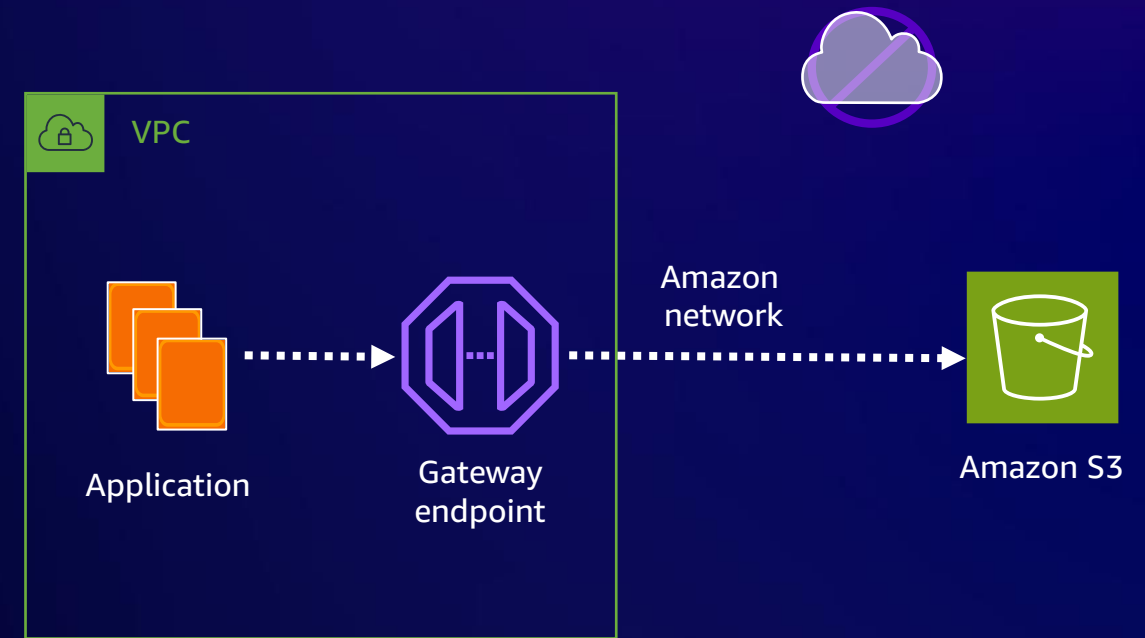


# Accessing Amazon S3 from a VPC using gateway endpoints

## Without gateway endpoint



## With gateway endpoint



- Simple network architecture
- Control access via endpoint policy
- Traffic remains on Amazon network
- Gateway endpoints are free

# Gateway endpoints: Based on the **routing table** in your VPC

search : s3 × Add filter

1 to 2 of 2

| Service Name                                                | Owner  | Type      |
|-------------------------------------------------------------|--------|-----------|
| <input checked="" type="radio"/> com.amazonaws.us-east-1.s3 | amazon | Gateway   |
| <input type="radio"/> com.amazonaws.us-east-1.s3            | amazon | Interface |

VPC\* vpc-c8b3a8b2 ▼ ↺ ℹ

Configure route tables

A rule with destination **pl-63a5400a (com.amazonaws.us-east-1.s3)** and a target with this endpoints' ID (e.g. vpce-12345678) will be added to the route tables you select below.

Subnets associated with selected route tables will be able to access this endpoint.

rtb-7cc3a502 ×

| Route Table ID                                   | Main | Associated With |
|--------------------------------------------------|------|-----------------|
| <input checked="" type="checkbox"/> rtb-7cc3a502 | Yes  | 6 subnets       |

aws

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# Gateway endpoints: Use standard Amazon S3 DNS

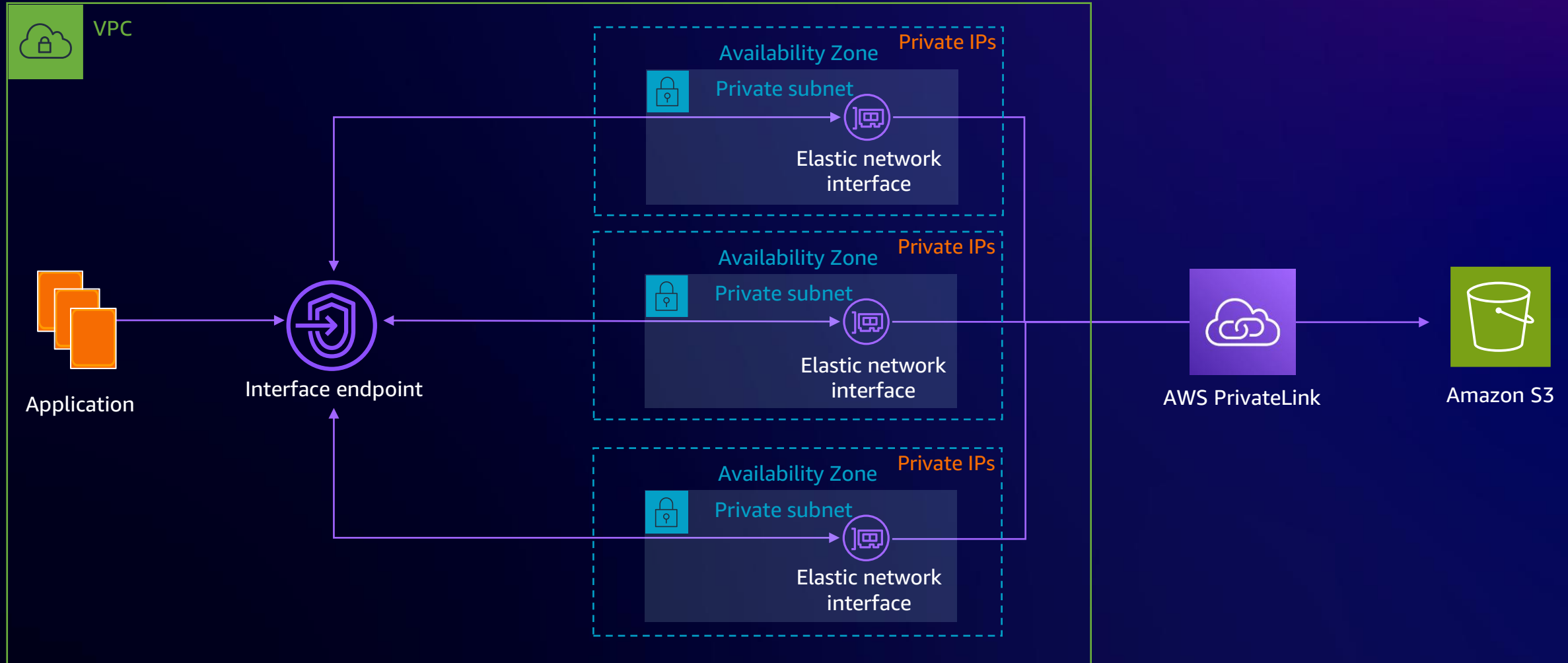
Standard S3 regional DNS names .....> S3 public IPs  
Example: s3.us-east-1.amazonaws.com

```
~ % nslookup s3.us-east-1.amazonaws.com
Server:      172.18.193.221
Address:     172.18.193.221#53

Non-authoritative answer:
Name:   s3.us-east-1.amazonaws.com
Address: 52.217.32.238

~ %
```

# Accessing Amazon S3 from a VPC using interface endpoints



# Interface endpoints: Based on the **private IPs** in your VPC

The screenshot displays the AWS Management Console interface. At the top, a search bar contains 's3' and a table lists VPC endpoints. Below this, a dropdown menu shows the selected VPC as 'vpc-c8b3a8b2'. A section titled 'Subnets' is highlighted with an orange border, showing a list of subnets with checkboxes for selection. The subnets are distributed across six availability zones in us-east-1.

| Service Name                                                | Owner  | Type      |
|-------------------------------------------------------------|--------|-----------|
| <input type="radio"/> com.amazonaws.us-east-1.s3            | amazon | Gateway   |
| <input checked="" type="radio"/> com.amazonaws.us-east-1.s3 | amazon | Interface |

VPC\* vpc-c8b3a8b2

Subnets subnet-4baf3106 subnet-68ea2137 subnet-7b7fb01d subnet-7503d654 subnet-c2ffedfc (+1 subnets)

| Availability Zone                                         | Subnet ID       |
|-----------------------------------------------------------|-----------------|
| <input checked="" type="checkbox"/> us-east-1a (use1-az4) | subnet-4baf3106 |
| <input checked="" type="checkbox"/> us-east-1b (use1-az6) | subnet-68ea2137 |
| <input checked="" type="checkbox"/> us-east-1c (use1-az1) | subnet-7b7fb01d |
| <input checked="" type="checkbox"/> us-east-1d (use1-az2) | subnet-7503d654 |
| <input checked="" type="checkbox"/> us-east-1e (use1-az3) | subnet-c2ffedfc |
| <input checked="" type="checkbox"/> us-east-1f (use1-az5) | subnet-ef8505e1 |

# Interface endpoints: Use endpoint-specific DNS

Regional endpoint-specific DNS names

`vpce-1a2b3c4d-5e6f.s3.us-east-1.vpce.amazonaws.com`



Private IPs of ENI in subnet 1

•  
•  
•

Private IPs of ENI in subnet 6

Zonal endpoint-specific DNS names

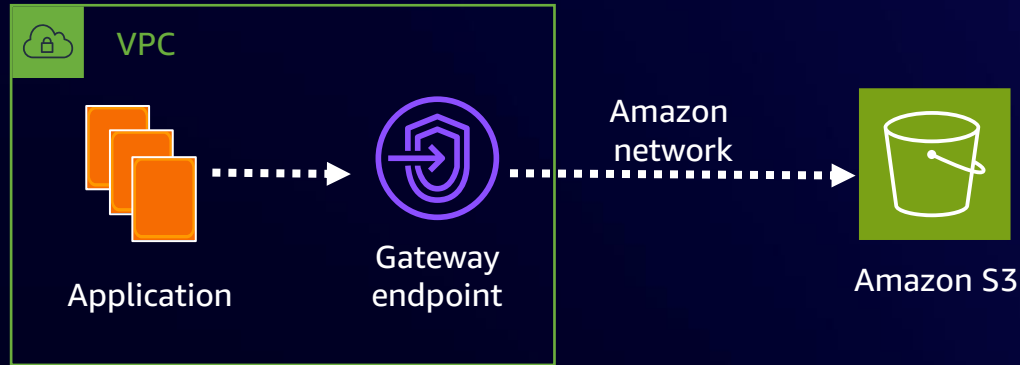
`vpce-1a2b3c4d-5e6f-us-east-1f.s3.us-east-1.vpce.amazonaws.com`



Private IP of ENI in subnet 1

# VPC endpoints: Gateway and interface

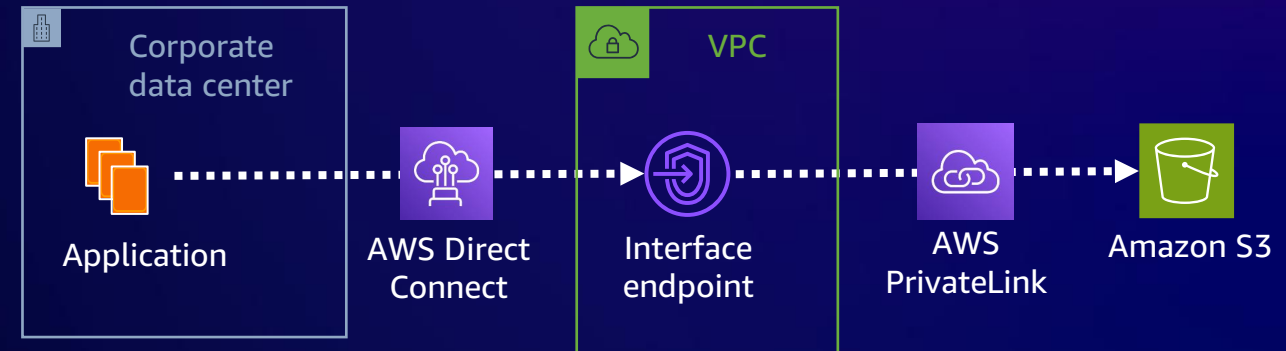
## Gateway endpoints



- Clients use public Amazon S3 IP addresses
- Routes based on routing table rule with target gateway endpoint
- No support for on-premises applications or cross-Region requests
- No additional charge

## Interface endpoints

Based on AWS PrivateLink

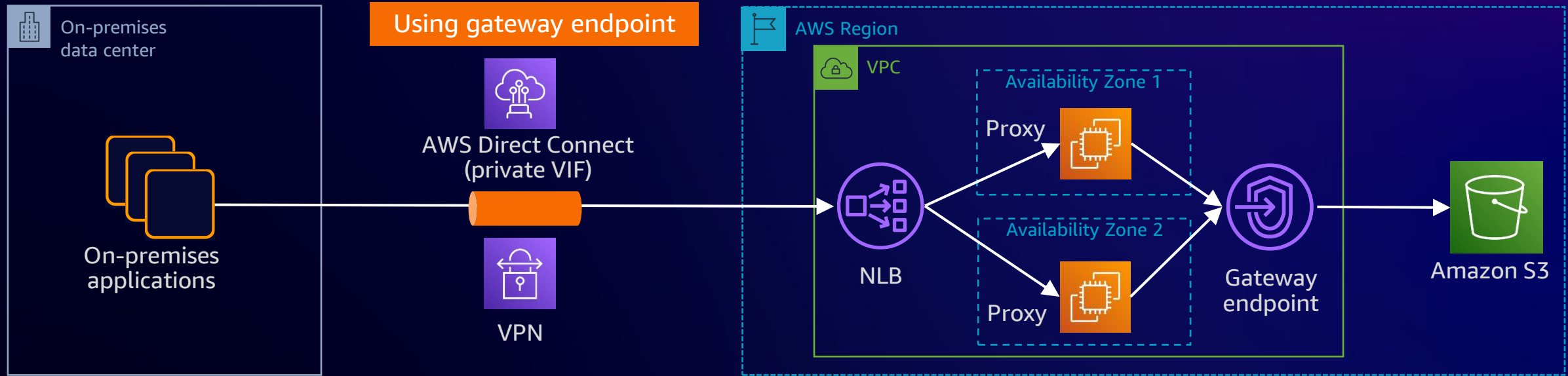


- Clients use IP addresses from their VPC
- Routes based on DNS entries for regional and ENI level
- Support for on-premises applications and cross-Region requests
- Paid feature

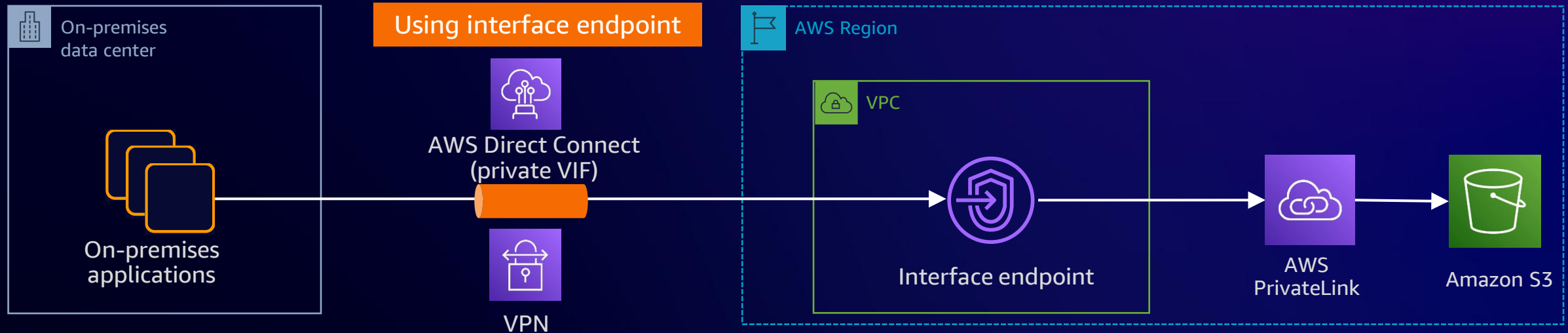
# Recent launches



# AWS PrivateLink for Amazon S3



# AWS PrivateLink for Amazon S3



# Private DNS name for interface endpoints for Amazon S3

## Problem statement

Endpoint-specific DNS name

for example, `vpce-1234.s3.us-east-1a.vpce.amazonaws.com`  
must be used by the consumer's application

Requires changes to the application

# Private DNS name for interface endpoints for S3

## Use Private DNS names for endpoint services

A private DNS name is specified by service provider during endpoint service configuration

### ▼ Additional settings

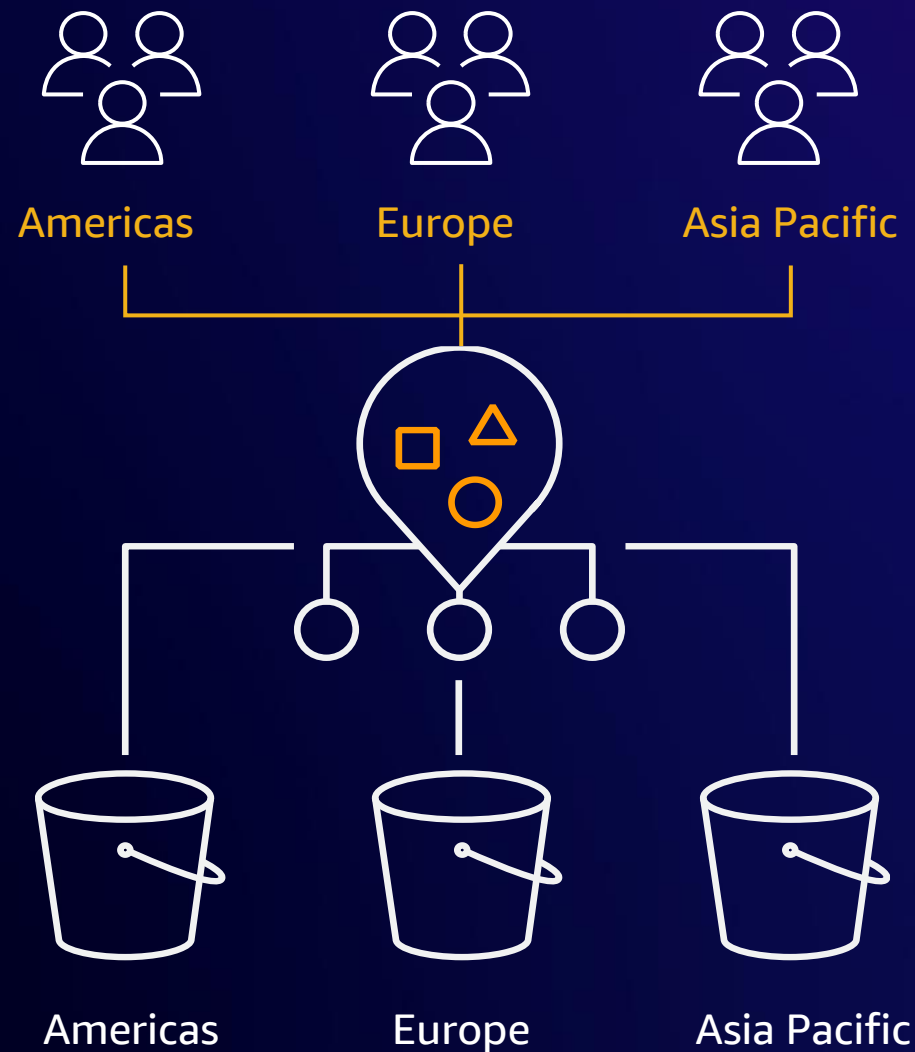
#### DNS name

☐ Enable DNS name [Info](#)

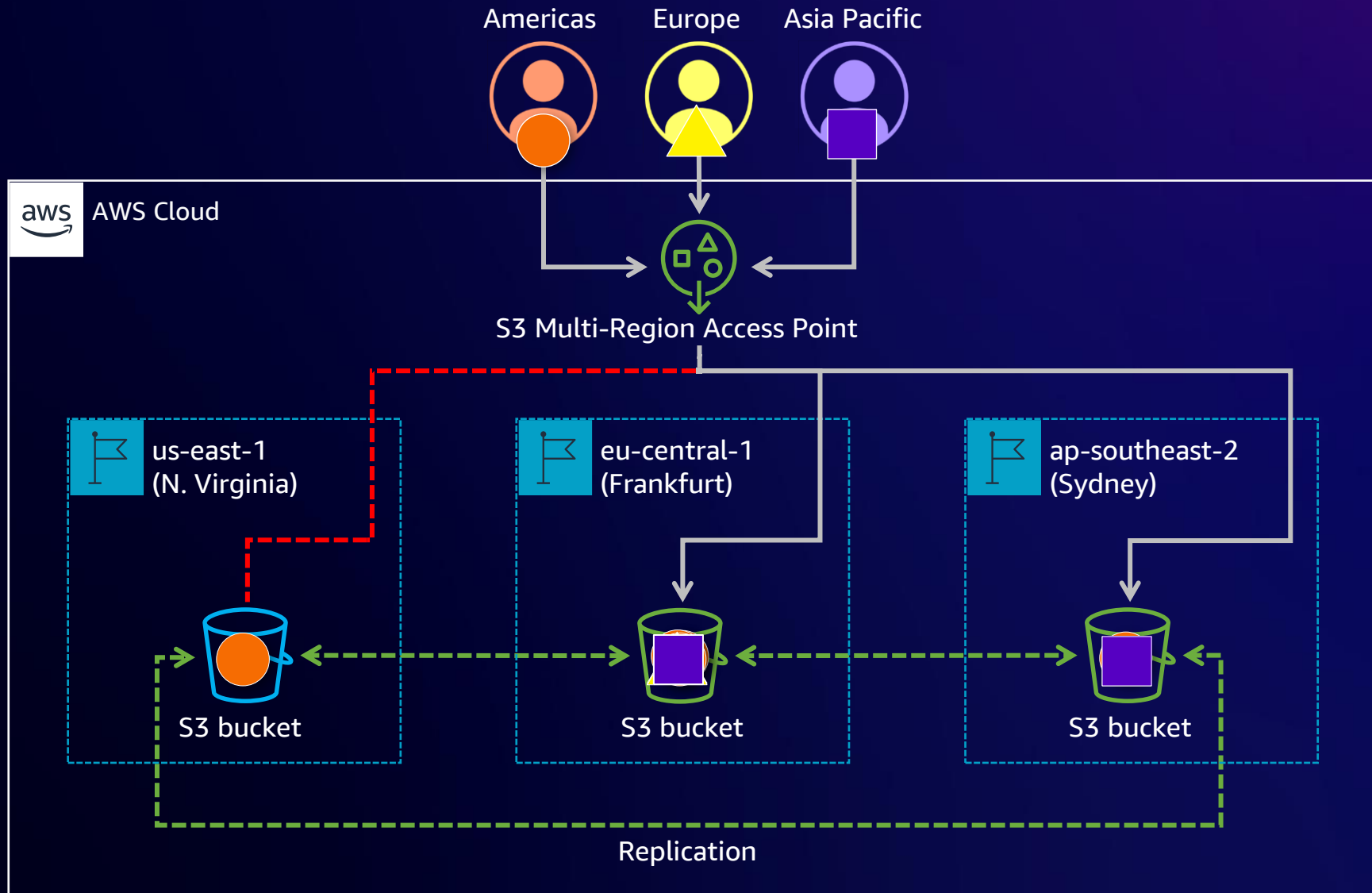
Associates a private hosted zone with the VPC that contains a record set that enables you to leverage Amazon's private network connectivity to the service while making requests to the service's default public endpoint DNS name. To use this feature, ensure that the attributes 'Enable DNS hostnames' and 'Enable DNS support' are enabled for your VPC.

☐ Enable private DNS only for inbound endpoint [Info](#)

# Amazon S3 Multi-Region Access Points



# S3 Multi-Region Access Points failover controls

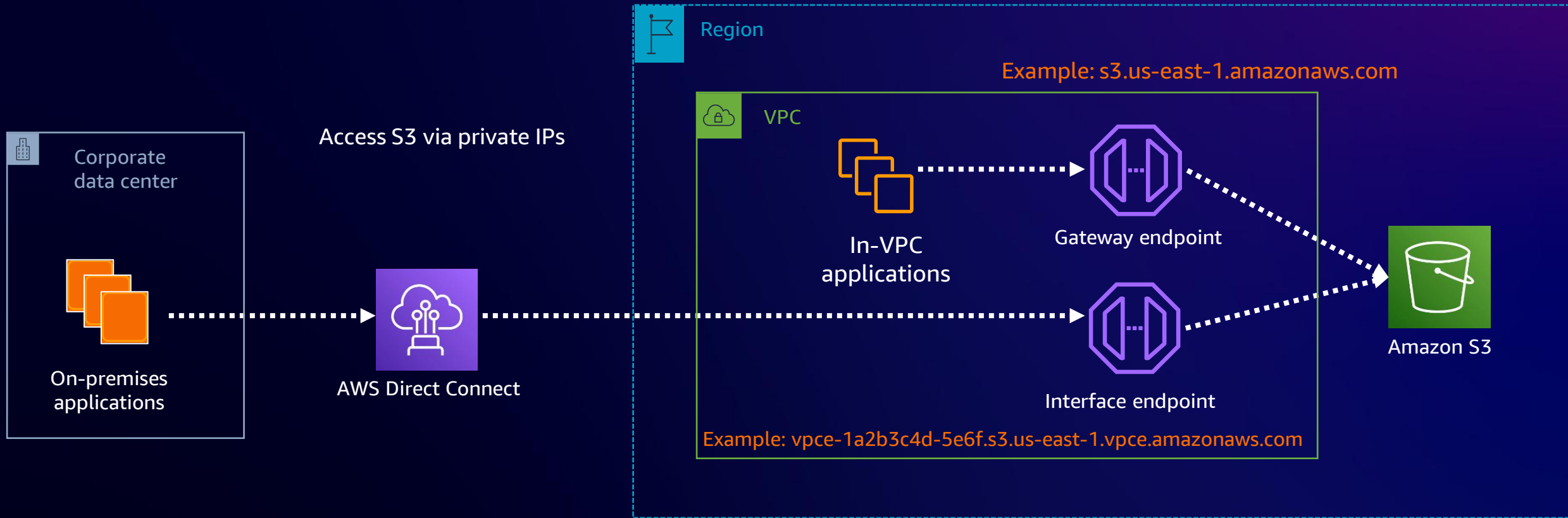


# Advanced use cases

Access Amazon S3 from:

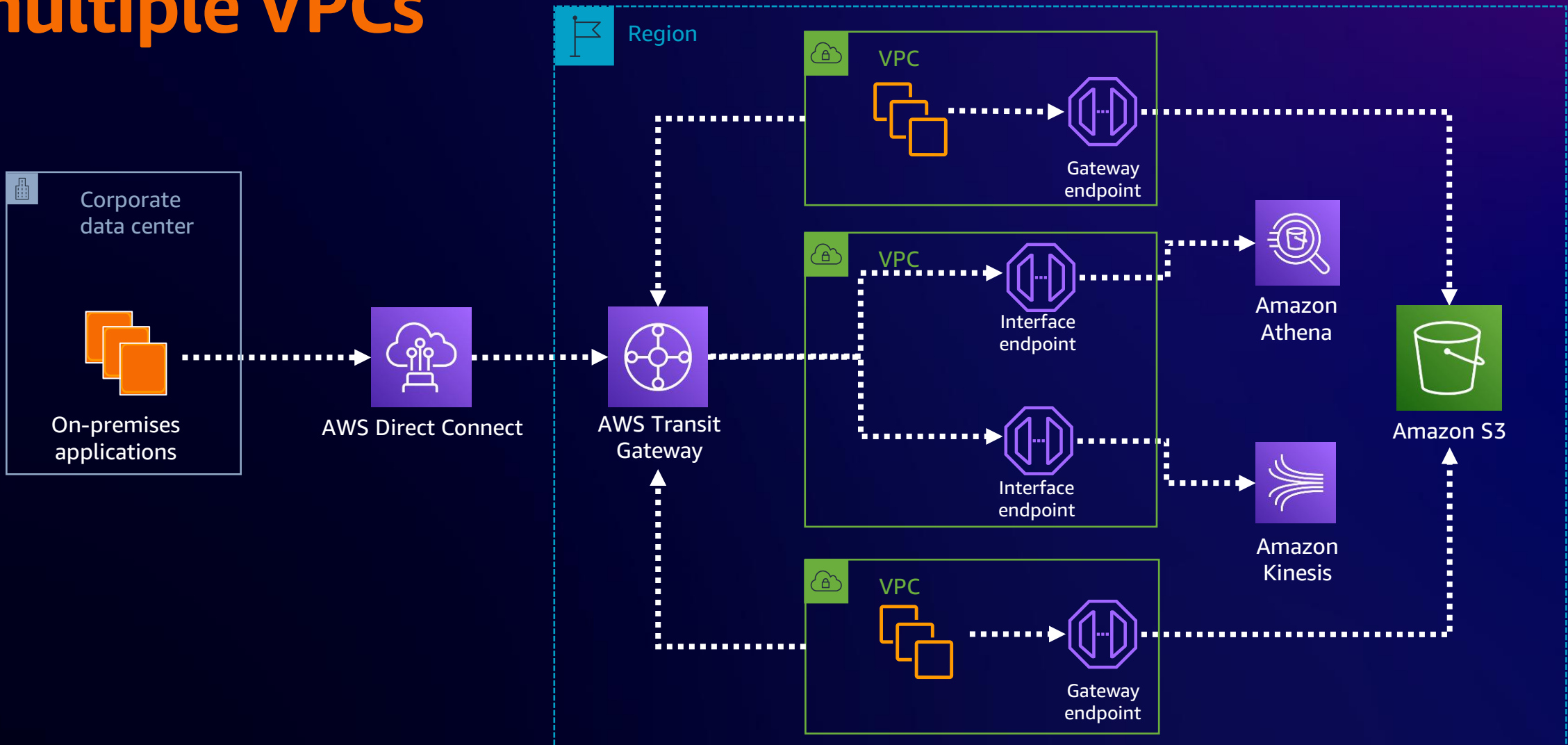
- On premises
- On premises and multiple VPCs
- A VPC in a different Region than the Amazon S3 bucket
- Using an S3 Multi-Region Access Point from VPC, on premises, and internet

# Case 1: Access Amazon S3 from **on premises**

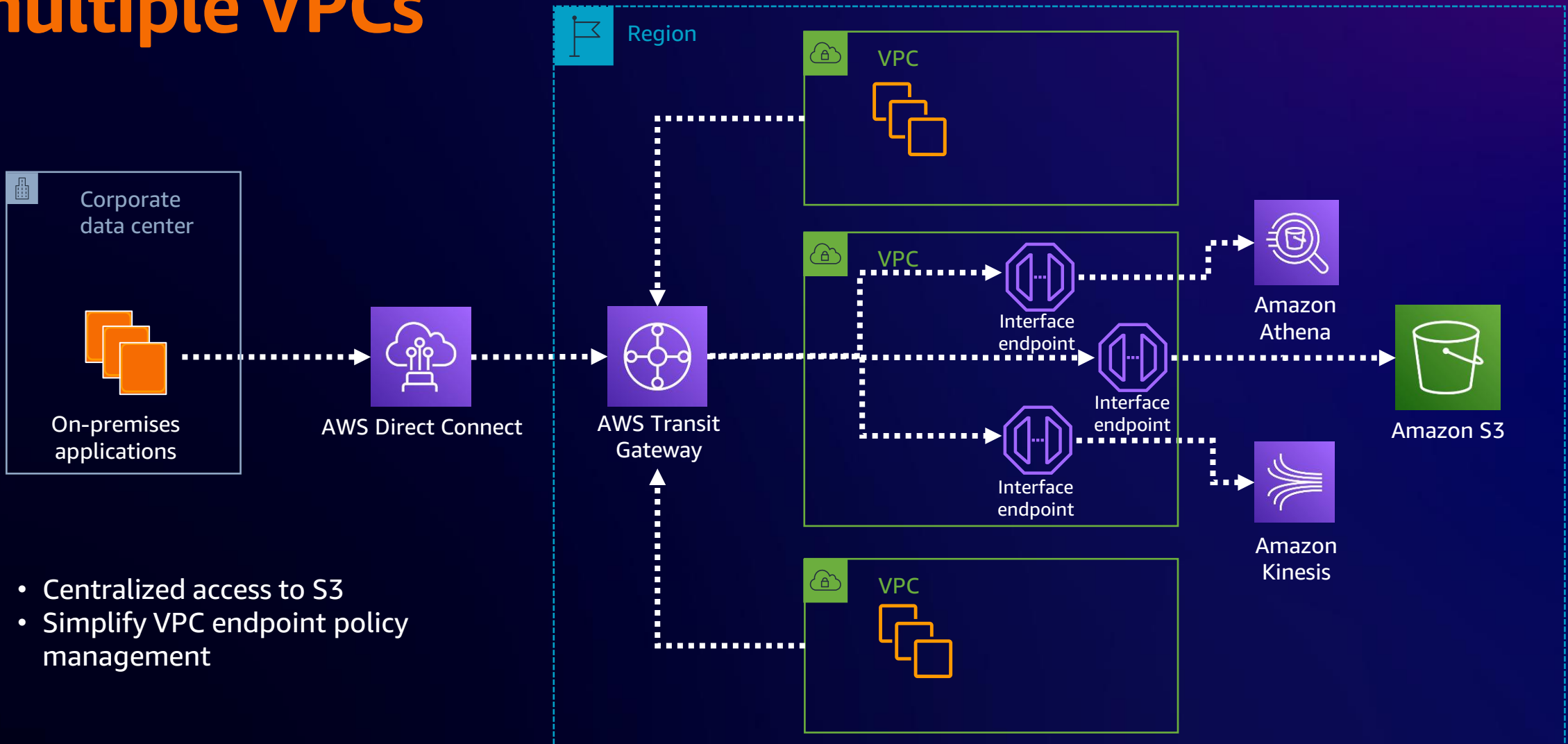


- **Optimize** cost
- **Leverage** existing VPC configuration with Direct Connect

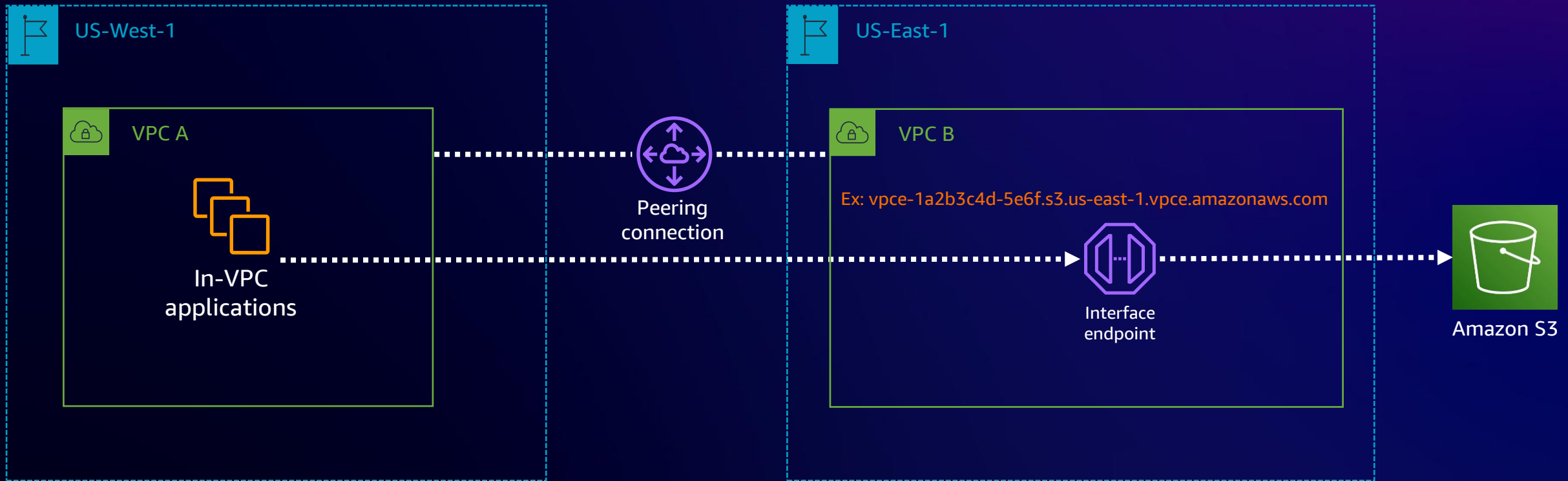
# Case 2: Access Amazon S3 from **on premises and multiple VPCs**



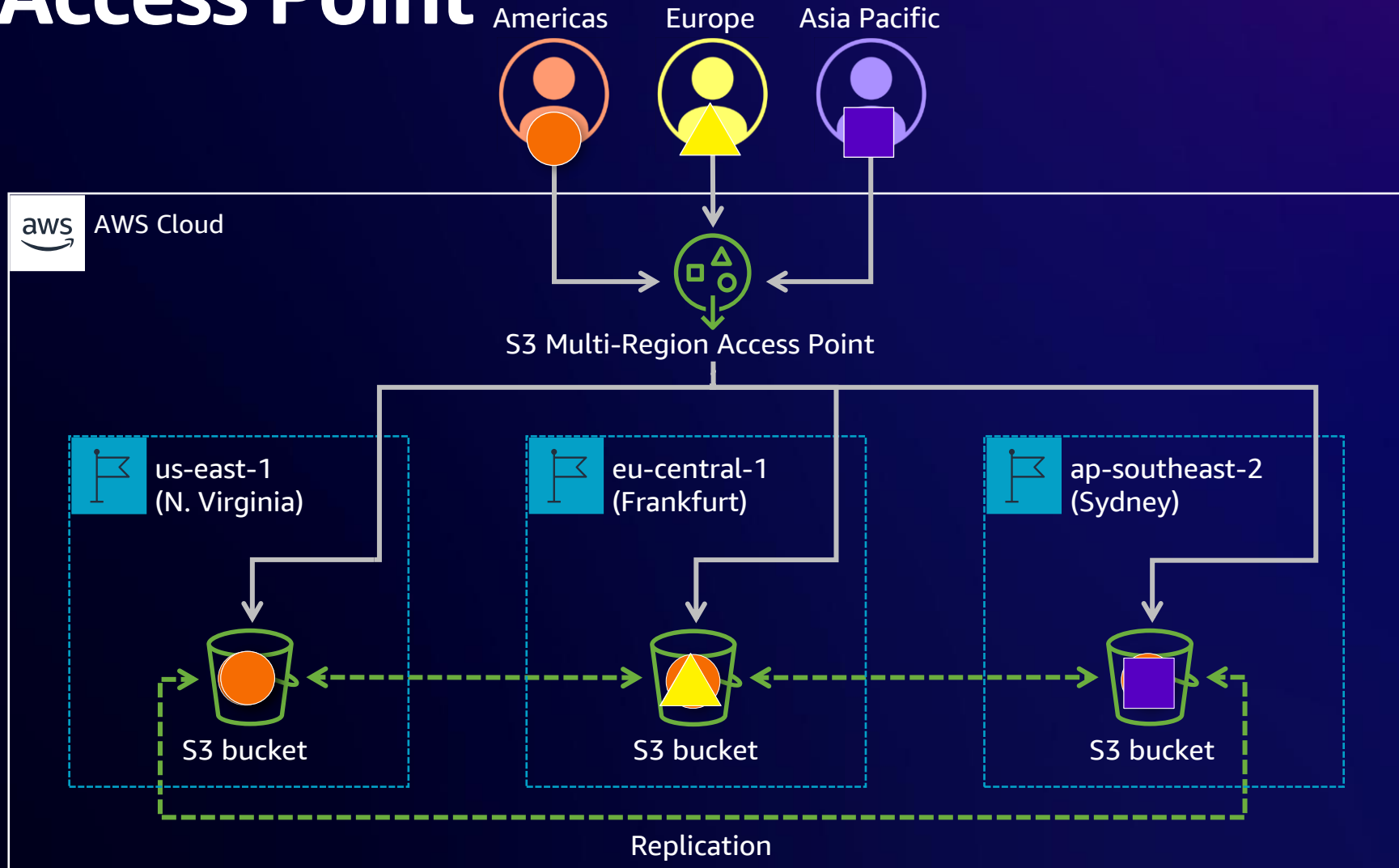
# Case 2: Access Amazon S3 from **on premises and multiple VPCs**



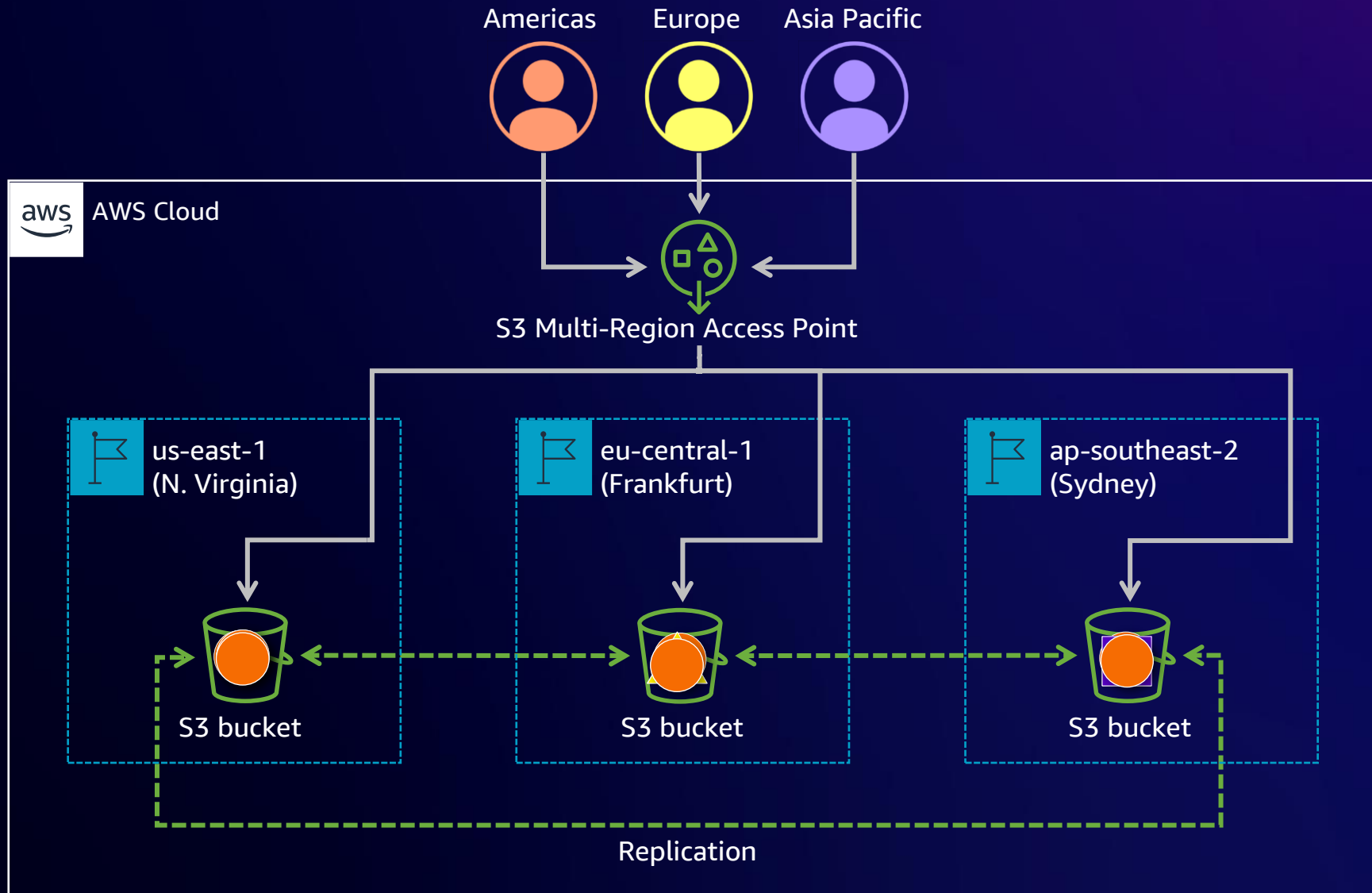
# Case 3: Accessing Amazon S3 from **cross-Region**



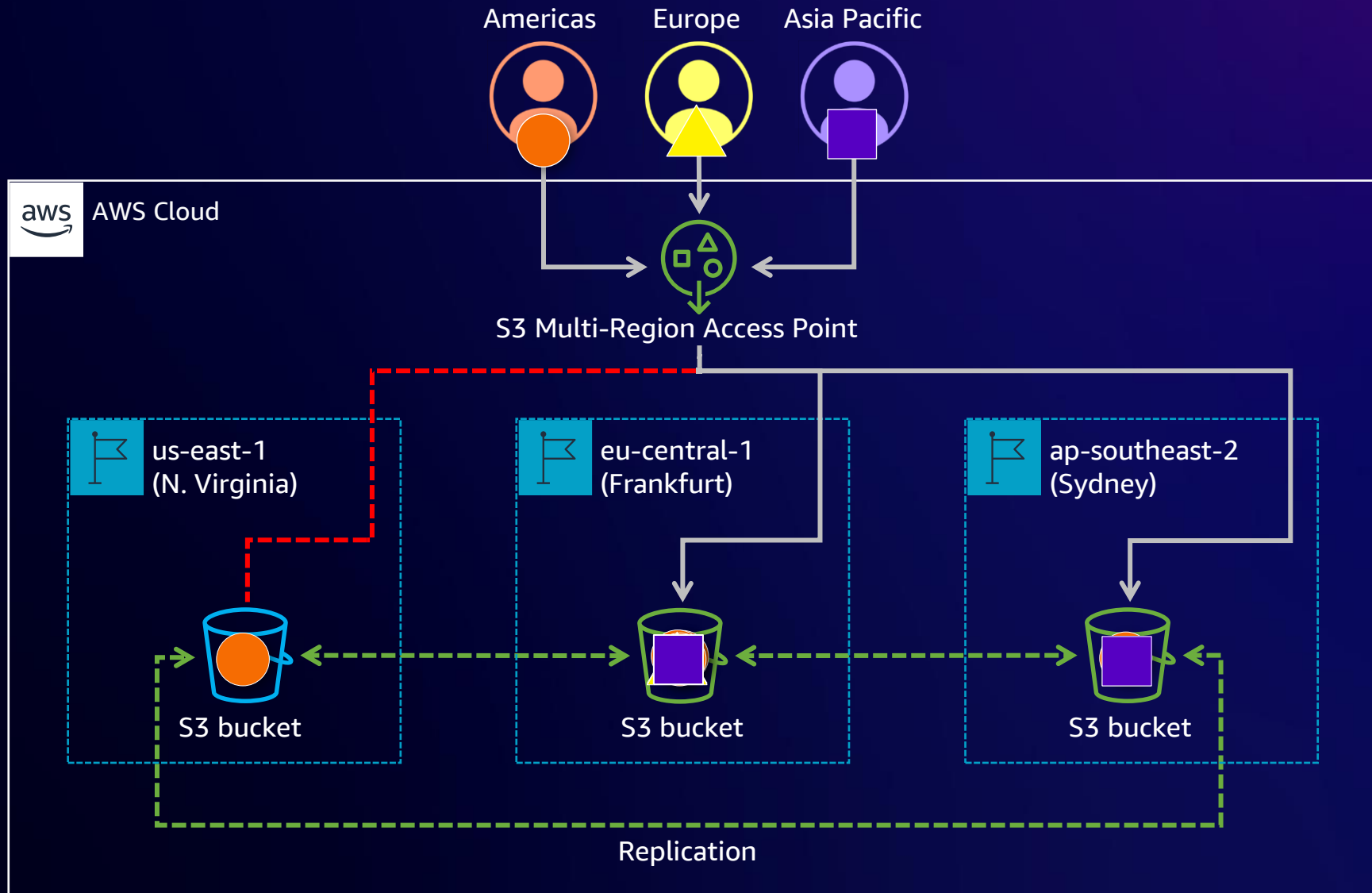
# Case 4: Accessing Amazon S3 using an S3 Multi-Region Access Point



# Active-active global storage



# S3 Multi-Region Access Points failover controls



# Recap



# Recap

- VPC endpoints provide secure access to Amazon S3 over the Amazon network without the need to configure an internet gateway or NAT devices
- Use gateway endpoints to connect from within VPC to Amazon S3 and use interface endpoints over AWS PrivateLink to connect from on premises to Amazon S3
- Use private DNS names for AWS PrivateLink for Amazon S3 while ensuring you're taking the cost-optimized network path for requests from within the VPC and on premises
- Maintain business continuity with S3 Multi-Region Access Points failover controls during Regional traffic disruptions
- Ways to securely connect from on premises, multiple VPCs, both, across Regions, and using MRAP

# Thank you!

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Please complete the session  
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